CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
1st Semester of B. Pharm. Examination
University Theory Examination January 2018
PH115 / PH124 Pharmaceutical Engineering-I

Date: 09/01/2018, Tuesday **Time: 10:00 a.m. to 1:00 p.m.** **Maximum Marks: 80**

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. 5 Lb = _______g
[A] 2267
[B] 2167
[C] 2230
[D] 2200

2. 2 ft = _______ Inches
[A] 12
[B] 36
[C] 24
[D] 10

3. Treated soda lime glass is also known as
[A]Type I
[B]Type II
[C]Type III
[D]Type IV

4. Tragacanth is showing ________ type of flow
[A] Plastic
[B] Dilatant
[C] Newtonian
[D] Pseudo Plastic

5. ____________ is the example of material indicating dilatants flow
[A] Butter
[B] Kaolin (12%) in water
[C] Methyl Cellulose
[D] Acacia

6. As temperature increases, viscosity of gas __________
[A] Decreases
[B] Increases
[C] No effect
[D] 1st increases and then decreases

7. As temperature increases, viscosity of fluid__________
[A] Decreases
[B] Increases
[C] No effect
[D] 1st increases and then decreases

8. For streamline flow, N_{Re} is __________
[A] > 4000
[B] <2100
[C] = 4000
[D] >2100

9. Foe critical flow, N_{Re} is __________
[A] 2100-4000
[B] 2200-4100
[C] 2200
[D] 2000-3000

10. _____________ is unit of viscosity
[A] Poise
[B] Dalton
[C] Kilo Dalton
[D] Dynes

11. Red colored pipe is used for handling _____________
[A] Fire fluids
[B] Gas
[C] Semisolids
[D] Gas and semisolids

12. 1 poise =_________ centipoise
[A] 10
[B]100
[C]1000
[D]10000

13. 500 cm = _____________ inches
[A] 300
[B] 110
[C] 100
[D] 198

14. Bernoulli's principle is valid for _________________ fluids.
[A] Incompressible
[B] Compressible
[C] Expandable
[D] Nonexpendable

15. 1Atmospeheric pressure = ____________mmHg
[A] 800
[B] 250
[C] 850
[D] 760

16. 10 °C =__________ ° F temperature.
[A] 52
[B] 55
[C] 50
[D] 60

17. The unit of Force (F) in MKS system is_________.
[A] Dyne
[B] Newton
[C] Kg
[D] Dalton

18. Which one of the following metal is used to improve corrosion resistance property of Carbon steel?
[A] Chromium
[B] Nickel
[C] Silicon
[D] Molybdenum

19. Fourier's law is applicable when
[A] Mode of Heat Transfer is conduction + Radiation
[B] Mode of Heat transfer by conduction
[C] Mode of Heat transfer by convection
[D] Mode of Heat transfer by Radiation

20. Body that radiates maximum possible quantity of energy is
[A] White body
[B] Black body
[C] Grey body
[D] Blue Body

SECTION – II

Q 2	Attempt any **TWO** of the following	
A	State the various types of fluid flow and give detail of them in brief.	**05**
B	Explain the Reynold's Experiment for fluid flow.	**05**
C	Define: Stoichiometry and unit operation. Give classification of unit operation.	**05**
Q 3	Attempt any **TWO** of the following	
A	Explain the phase rule.	**05**
B	Enlist the various types of manometer and explain any one in detail.	**05**
C	Enlist the various mode of heat transfer. Explain any one in deial.	**05**

SECTION – III

Q 4	Attempt any **FOUR** of the following	
A	Give principle of mass transfer. Describe influences of mass transfer on unit operations.	**05**
B	Describe factors affecting transfer of mass from solid to a fluid with diagrams.	**05**
C	Give a detail note on centrifugal pump.	**05**
D	Write principle, construction and working of belt conveyer.	**05**
E	Discuss theories of corrosion.	**05**
F	Write a short note on utility of glass and stainless steel in pharmaceutical industry.	**05**

Q 5	Attempt any **FOUR** of the following	
A	(i) Describe color coding of pipelines.	**03**
	(ii) Give classification of solids transport system.	**02**
B	Discuss uses of forklifts and pallets.	**05**
C	Write classification of materials for plant construction. Mention applications of plastic in pharmaceutical industries.	**05**
D	Define corrosion. Write a note on prevention of corrosion in a pharmaceutical plant.	**05**
E	Give classification of packaging material used in pharmaceutical industries. Describe qualities of packaging material.	**05**
F	Explain various factors affecting selection of material of plant construction.	**05**